B BELLOMASSI

API AND INTEGRATION

API Specification (Overview and Conventions)

Bellomassi Order-to-Delivery Management Platform

Document ID BM-API-001

Document control

Field	Value
Document title	API Specification (Overview and Conventions)
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> **Status labels used in this document:** VERIFIED (confirmed in the current source code or existing documents) · PROTOTYPE (works in the React/LocalStorage prototype with mock or simulated behaviour) · PROPOSED (recommended production design, not built) · PENDING CONFIRMATION (needs a client decision) · NOT IMPLEMENTED · NOT VERIFIED.

Purpose

Define the conventions of the proposed Bellomassi REST API and separate clearly what exists today (frontend services, mock behaviour) from what is planned.

Scope

REST API v1 for web, driver and customer clients; WebSocket channels; webhooks. Everything in this folder except section 1 is PROPOSED.

Definitions

Other terms: see **BM-GLO-001** (Project Glossary).

References

- docs/INTEGRATION_GATEWAY.md — gateway contract for email, OpenWA, OCR, AI (exists as a contract only)
- Driver App GPS Implementation Guide §11 — example driver endpoints

- BM-API-018_openapi_proposed.yaml — OpenAPI 3.0 file (validated with openapi-spec-validator)
- **BM-API-002** (API Endpoint Catalogue)
- **BM-API-019** (Integration Specification (External Systems))
- **BM-ARC-001** (High-Level Design (HLD))
- **BM-SEC-003** (Authentication, Password, MFA and Session Management Design)
- **BM-DAT-002** (Proposed Production Database Design)

1. What exists today

Layer	Status	Description
Frontend functionality	VERIFIED	84 browser service modules implement every workflow (e.g. `orderService.submitOrder`, `approvalService.approveOrder`, `deliveryService.confirmDelivery`).
Mock API behaviour	PROTOTYPE	Services return `{ ok, record, error, errors }` synchronously from LocalStorage; simulated providers for GPS, OTP, email, WhatsApp, OCR, AI.
Integration gateway contract	NOT VERIFIED	Optional HTTP contract (`VITE_INTEGRATION_GATEWAY_URL`) for email, OpenWA, OCR, AI and ML; no gateway is deployed.
REST API	PROPOSED	This specification — 140 endpoints in 15 domains (**BM-API-002** (API Endpoint Catalogue)).
Production integrations	PROPOSED	Maps / Routes, SMS / WhatsApp, email, OCR, SAP (**BM-API-019** (Integration Specification (External Systems))).

2. Conventions

Topic	Convention
Base URL	`https://<host>/api/v1` (versioned in the path; breaking changes → v2)
Format	JSON (UTF-8); `multipart/form-data` for files; ISO-8601 dates; times UTC with offset; money as decimal strings
Authentication	`Authorization: Bearer <JWT access token>` (15 min); refresh token rotation; MFA for privileged roles
Authorisation	Permission key per endpoint (same 110 keys as the prototype) + object-level data scope
Pagination	`?page=1&page_size=25` → `{count, next, previous, results[]}` (max page size 200)
Filtering / sorting	Query parameters (`status`, `from`, `to`, `search`, `ordering=-created_at`)
Idempotency	`Idempotency-Key` header required on submit, approve, confirm, complete, write-off; repeated keys return the first result
Concurrency	`version` field (optimistic locking) on orders, dispatches, stock; `409 Conflict` on mismatch
Errors	`{ "code": "ORDER_NOT_EDITABLE", "message": "…", "errors": { "field": "message" } }`
Status codes	200 / 201 / 202 / 204; 400 validation; 401 unauthenticated; 403 forbidden; 404 not found or out of scope; 409 state conflict; 413 / 415 file; 422 business rule; 429 rate limit; 503 provider unavailable
Rate limiting	Per user and per IP; stricter on auth, OTP and password endpoints
Real-time	WebSocket `wss://<host>/ws/` channels: `tracking.dispatch.<id>`, `tracking.order.<id>` (customer, own order only), `monitor.orders`, `notifications.user.<id>`
Webhooks	Provider callbacks signed (HMAC) and verified; replay-protected

Topic	Convention
Documentation	OpenAPI 3.0 generated by drf-spectacular; draft file BM-API-018

3. Error message reuse

The prototype's user messages (e.g. "Only draft orders can be deleted. Submitted orders are kept for the audit trail.") are the reference texts; each business rule in **BM-SYS-007** (Business Rules Catalogue) becomes an error `code`.

4. Domain designs

Domain	Document
Authentication	**BM-API-003** (Authentication API Design)
Users, roles, audit	**BM-API-004** (User API Design)
Customers	**BM-API-005** (Customer API Design)
Products, hubs, zones	**BM-API-006** (Product API Design)
Orders	**BM-API-007** (Order API Design)
Approval and consolidation	**BM-API-008** (Approval API Design)
Dispatch	**BM-API-009** (Dispatch API Design)
Driver	**BM-API-010** (Driver API Design)
Delivery, OTP, tracking	**BM-API-011** (Delivery API Design)
Inventory, scrap, replenishment	**BM-API-012** (Inventory API Design)
Vehicles and VMS	**BM-API-013** (Vehicle API Design)
Documents and OCR	**BM-API-014** (Document API Design)
Invoices	**BM-API-015** (Invoice API Design)
Notifications, messages, calls	**BM-API-016** (Notification API Design)
Dashboards and reports	**BM-API-017** (Reporting API Design)

Revision history

Version	Date	Author	Description
1.0	2026-11-12	Documentation team	Initial draft prepared from analysis of the bellomassi-premium-ui source code and existing project documents.